

本次课内容

3.2 动量算符和角动量算符

- § 3.3 中心力场中的运动，氢原子 (1) 
(涉及较复杂的征值问题)

预习思考题

- 为什么动量是连续谱，角动量却是离散谱？（角动量为什么是量子化的？）
- 角动量每个分量都是量子化的吗？
- 氢原子模型中为什么引入约化质量？氢原子模型中动量算符 $\vec{p} = -i\hbar\nabla$ 代表什么物理量？原子内部能级是电子运动的能量吗？
- 氢原子能量量子化是因为角动量量子化吗？

回顾：量子力学的基本假设

物理状态：用波函数表示 $\psi(\vec{r}, t)$

力学量：用力学量算符表示 $\hat{F}(\vec{r}, \vec{p})$

- 测量假设：力学量算符（厄密算符）本征值集合对应力学量实际测量值集合。

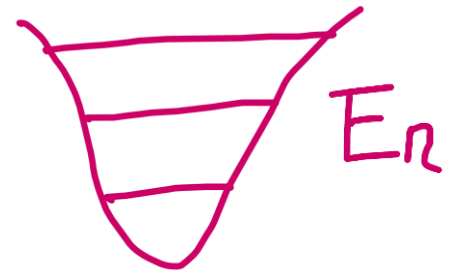
$$\hat{F}\psi_\lambda = \lambda\psi_\lambda$$

一般状态下力学量的值不确定，其概率分布由波函数决定（将波函数按照力学量本征函数展开可求得）。F本征态下F的值确定。

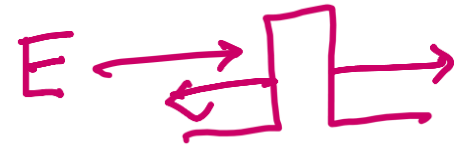
- 波函数满足薛定谔方程。

回顾：能量本征值问题

定态下能量确定 $H\psi_n = E_n\psi_n$



- 束缚态：势能决定能级
- 非束缚态（如势垒穿透）：能量连续。
势能决定透射率与反射率。
(能量E都是全空间的)



3.2 动量算符和角动量算符

一维动量本征函数和它的归一化

$$\hat{p} = -i\hbar \frac{d}{dx},$$

$$\hat{p} \psi_p = p \psi_p, \quad -i\hbar \frac{d\psi}{dx} = p\psi,$$

解是

$$C \exp\left(\frac{ipx}{\hbar}\right), \quad p \text{ 是任意实数} \\ \text{(连续谱)}$$

三维情况

$$\hat{\vec{p}} = -i\hbar\nabla, \quad -i\hbar\nabla\psi_{\vec{p}} = \vec{p}\psi_{\vec{p}},$$

按分量写出是

$$-i\hbar\frac{\partial\psi}{\partial x} = p_x\psi, \quad -i\hbar\frac{\partial\psi}{\partial y} = p_y\psi, \quad -i\hbar\frac{\partial\psi}{\partial z} = p_z\psi,$$

它们的解分别是

$$C_1\exp\left(\frac{ip_x x}{\hbar}\right), \quad C_2\exp\left(\frac{ip_y y}{\hbar}\right), \quad C_3\exp\left(\frac{ip_z z}{\hbar}\right),$$

三维动量本征函数

$$\psi_{\vec{p}}(\vec{r}) = \frac{1}{\sqrt{(2\pi\hbar)^3}} \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right).$$

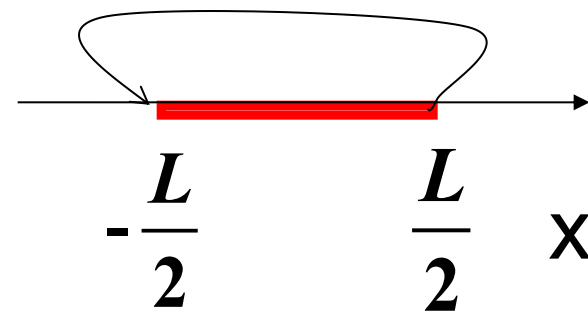
$\vec{p}$ 是任意实数向量
(连续谱)

$$\int_{\infty} \psi_{\vec{p}'}^*(\vec{r}) \psi_{\vec{p}}(\vec{r}) \cdot d\tau = \delta^3(\vec{p} - \vec{p}'),$$

*箱归一化

一维: $\psi_p(x) = C \exp\left(\frac{i}{\hbar} px\right)$ (动量本征态)

为了归一化, 人为设定条件:



$$\psi_p\left(\frac{L}{2}\right) = \psi_p\left(-\frac{L}{2}\right) \Rightarrow p = \frac{2\pi\hbar}{L} n \quad (n = 0, \pm 1, \pm 2, \dots)$$

归一化, 取 $C = \frac{1}{\sqrt{L}}$

$L \rightarrow +\infty$ 准连续

$$\int_{-L/2}^{L/2} \psi_p^*(x) \psi_{p'}(x) dx = \delta_{pp'}$$

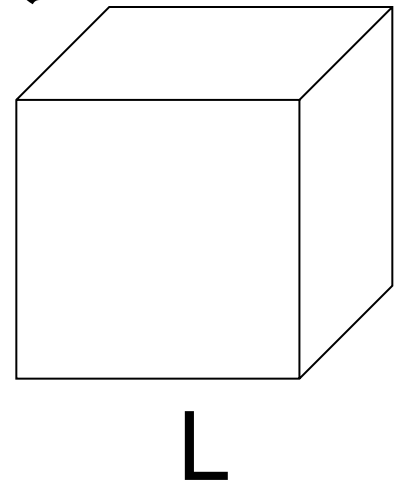
$$\psi(x, t) = \sum_p c_p(t) \psi_p(x) \Rightarrow \psi\left(-\frac{L}{2}, t\right) = \psi\left(\frac{L}{2}, t\right)$$

三维: $\psi_{\vec{p}}(\vec{r}) = \frac{1}{\sqrt{L^3}} \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right)$ (动量本征函数)

$p_x = \frac{2\pi\hbar}{L} n_x$ ($n_x = 0, \pm 1, \pm 2, \dots$) “装在无限大的箱子里”

$p_y = \frac{2\pi\hbar}{L} n_y$ ($n_y = 0, \pm 1, \pm 2, \dots$) ($L \rightarrow +\infty$)

$p_z = \frac{2\pi\hbar}{L} n_z$ ($n_z = 0, \pm 1, \pm 2, \dots$) 准连续



$$\int_{-\infty}^{\infty} \psi_{\vec{p}}(\vec{r}) \psi_{\vec{p}'}(\vec{r}) d^3\vec{r} = \delta_{\vec{p}, \vec{p}'}$$

一般波函数: $\psi(\vec{r}, t) = \sum_{\vec{p}} c_{\vec{p}}(t) \psi_{\vec{p}}(\vec{r})$

原始

$$\psi_{\vec{p}}(\vec{r}) = \frac{1}{\sqrt{(2\pi\hbar)^3}} \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right)$$

动量为连续谱

$$\int_{\infty} \psi_{\vec{p}'}^*(\vec{r}) \psi_{\vec{p}}(\vec{r}) \cdot d\tau = \delta^3(\vec{p} - \vec{p}'),$$

- 箱归一化可以得到正交归一的本征函数系，是一种常用处理方法（例如在固体物理、量子光学等领域常用）
- 原始本征函数（未归一化）仍然可以用。

箱归一化

$$\tilde{\psi}_{\vec{p}}(\vec{r}) = \frac{1}{\sqrt{L^3}} \exp\left(\frac{i}{\hbar} \vec{p} \cdot \vec{r}\right)$$

$$(L \rightarrow +\infty)$$

动量为准连续谱

$$\int_{\text{箱}} \tilde{\psi}_{\vec{p}'}^*(\vec{r}) \tilde{\psi}_{\vec{p}}(\vec{r}) d^3\vec{r} = \delta_{\vec{p}, \vec{p}'}$$

思考题：能不能将一维“箱归一化”的动量本征函数改为一维无限深势阱的定态波函数？

答案：不能！

（原因是一维无限深势阱的定态波函数不是动量本征函数。）

***例题：**以一维情况分析动量算符是否满足厄密性要求.

厄密性要求 $\int \psi^* (\hat{p} \phi) \cdot dx = \int (\hat{p} \psi)^* \phi \cdot dx$, 即

$$\int \psi^* \left(-i\hbar \frac{d}{dx}\right) \phi \cdot dx = \int \left(-i\hbar \frac{d}{dx} \psi\right)^* \phi \cdot dx,$$

$$\text{移项} \Rightarrow \int dx \left(\psi^* \frac{d\phi}{dx} + \frac{d\psi^*}{dx} \phi \right) = 0 \Rightarrow \int dx \frac{d}{dx} (\psi^* \phi) = 0$$

$$\Rightarrow \psi^* \phi \Big|_{-\infty}^{+\infty} = 0 \Rightarrow \psi^*(+\infty)\phi(+\infty) = \psi^*(-\infty)\phi(-\infty)$$

(1) 若假设无穷远处波函数为零, 则厄密性满足。

(2) 若满足箱归一化条件, 即

$$\psi^*(L/2)\phi(L/2) = \psi^*(-L/2)\phi(-L/2) \quad (L \rightarrow \pm\infty)$$

则满足厄密性也满足。

3. 角动量算符及其本征函数

- 角动量算符的定义是:

$$\vec{\hat{L}} = \vec{\hat{r}} \times \vec{\hat{p}} = -i\hbar \vec{\hat{r}} \times \nabla,$$

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y$$

$$\hat{L}_x = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right), \dots$$

$$\hat{L}_y = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right), \dots$$

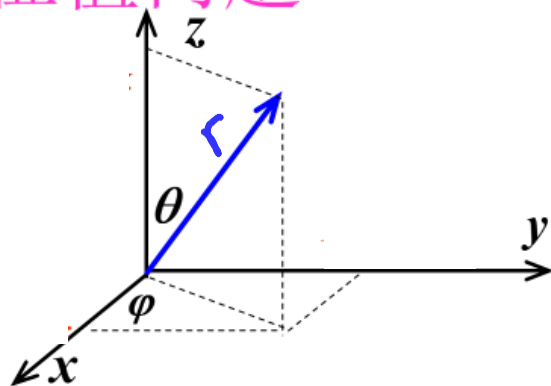
$$\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right).$$

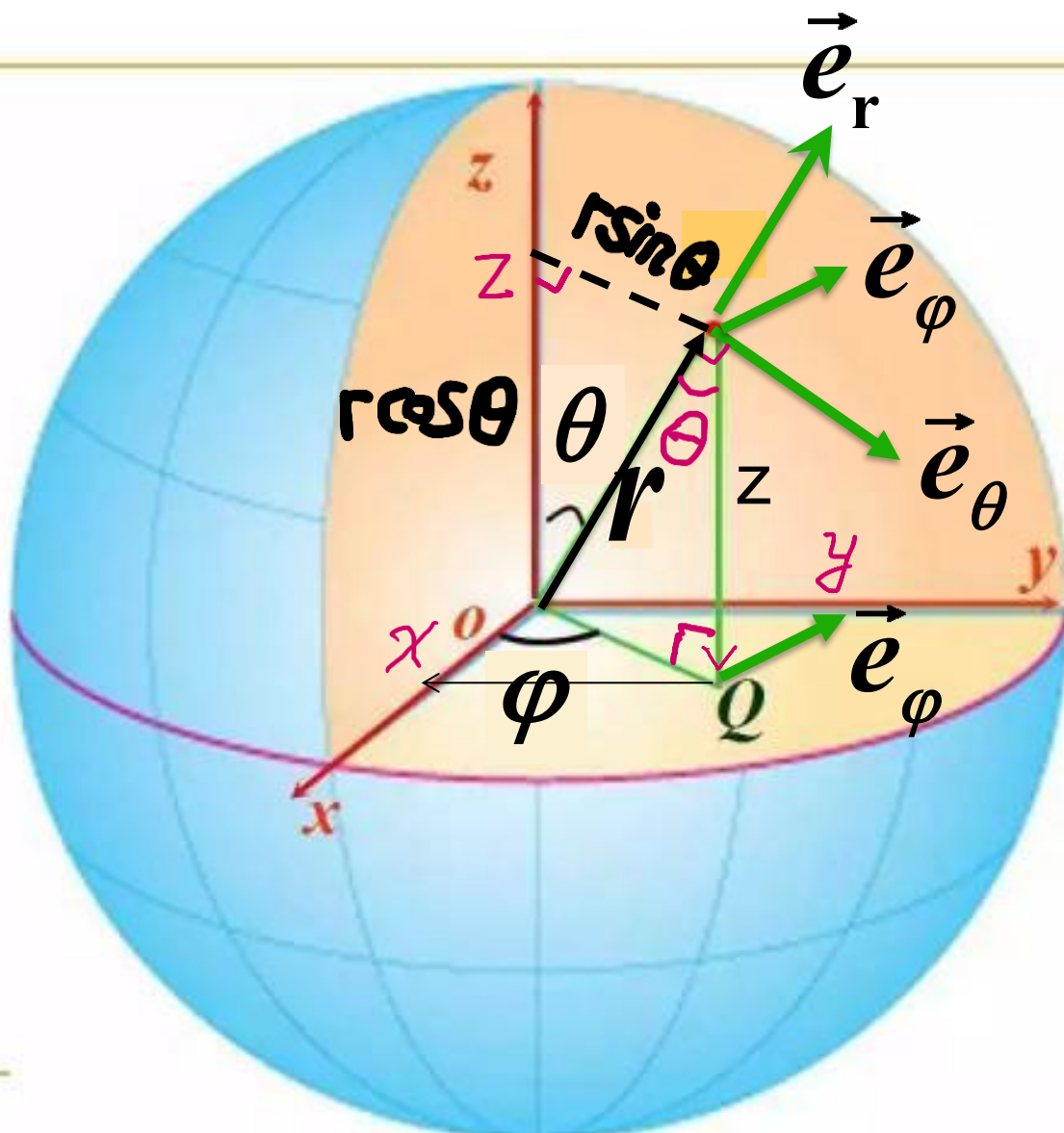
也可以由行列式构造

方向矢量(单位向量)

$$\vec{\hat{L}} = \vec{\hat{r}} \times \vec{\hat{p}} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \hat{x} & \hat{y} & \hat{z} \\ \hat{p}_x & \hat{p}_y & \hat{p}_z \end{vmatrix}$$

变换成球坐标求解
本征值问题





$$z = r \cos \theta$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

微分线元

$$dr, \quad r d\theta,$$

$$r \sin \theta d\phi$$

$$r \in [0, \infty), \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi)$$

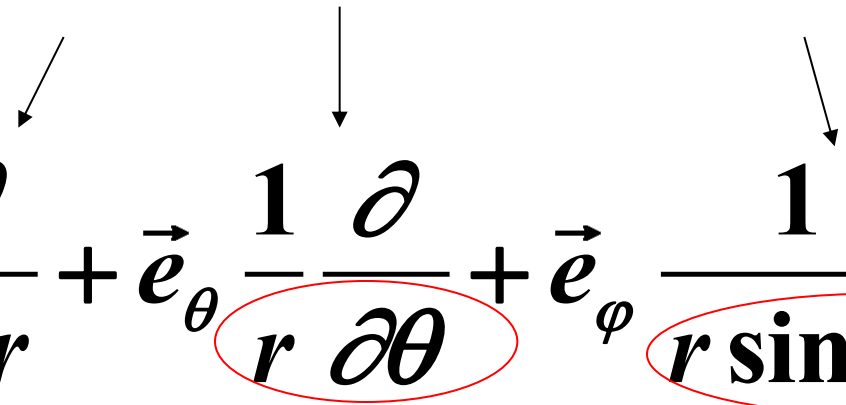
梯度

直角坐标: $\nabla = \vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z}$

线元 dx 、 dy 、 dz

线元 dr 、 $r d\theta$ 、 $r \sin \theta d\varphi$

球坐标:

$$\nabla = \vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_\varphi \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi}$$


利用此公式给出角动量（比教材上简单）

用球坐标表示角动量算符



$$(\vec{p} = -i\hbar\nabla)$$

$$\vec{L} = \vec{r} \times \vec{p} = -i\hbar\vec{r} \times \nabla = -i\hbar r \vec{e}_r \times \nabla$$

$$= -i\hbar r \vec{e}_r \times \left(\vec{e}_r \frac{\partial}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{e}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \right)$$

$$\vec{e}_r \times \vec{e}_r = 0, \vec{e}_r \times \vec{e}_\theta = \vec{e}_\phi, \vec{e}_r \times \vec{e}_\phi = -\vec{e}_\theta$$

$$\vec{L} = -i\hbar \left(\vec{e}_\phi \frac{\partial}{\partial \theta} - \vec{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right)$$

如何写成直角坐标？ *

这是球坐标形式。
如何进一步写出
 L_x 、 L_y 、 L_z ？

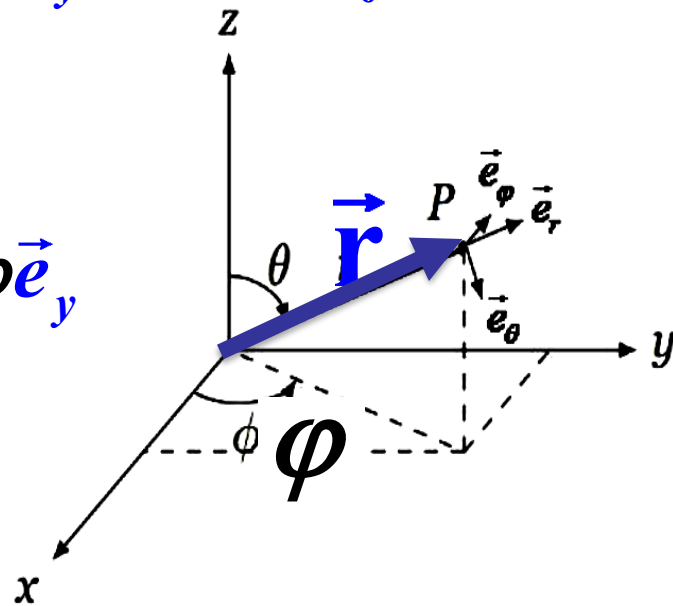
*附：将球坐标单位向量用直角坐标表示

$$\vec{r} = r \sin \theta \cos \varphi \vec{e}_x + r \sin \theta \sin \varphi \vec{e}_y + r \cos \theta \vec{e}_z \Rightarrow$$

$$\vec{e}_r = \frac{\vec{r}}{r} = \sin \theta \cos \varphi \vec{e}_x + \sin \theta \sin \varphi \vec{e}_y + \cos \theta \vec{e}_z$$

$$\vec{e}_\varphi = \frac{\frac{\partial \vec{r}}{\partial \varphi} d\varphi}{\left| \frac{\partial \vec{r}}{\partial \varphi} \right| d\varphi} = -\sin \varphi \vec{e}_x + \cos \varphi \vec{e}_y$$

$$\vec{e}_\theta = \frac{\frac{\partial \vec{r}}{\partial \theta} d\theta}{\left| \frac{\partial \vec{r}}{\partial \theta} \right| d\theta} = \cos \theta \cos \varphi \vec{e}_x + \cos \theta \sin \varphi \vec{e}_y - \sin \theta \vec{e}_z$$



$$\vec{L} = -i\hbar(\vec{e}_\varphi \frac{\partial}{\partial \theta} - \vec{e}_\theta \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi})$$

$$\vec{e}_\varphi = -\sin \varphi \vec{e}_x + \cos \varphi \vec{e}_y$$

$$\vec{e}_\theta = \cos \theta \cos \varphi \vec{e}_x + \cos \theta \sin \varphi \vec{e}_y - \sin \theta \vec{e}_z$$

于是

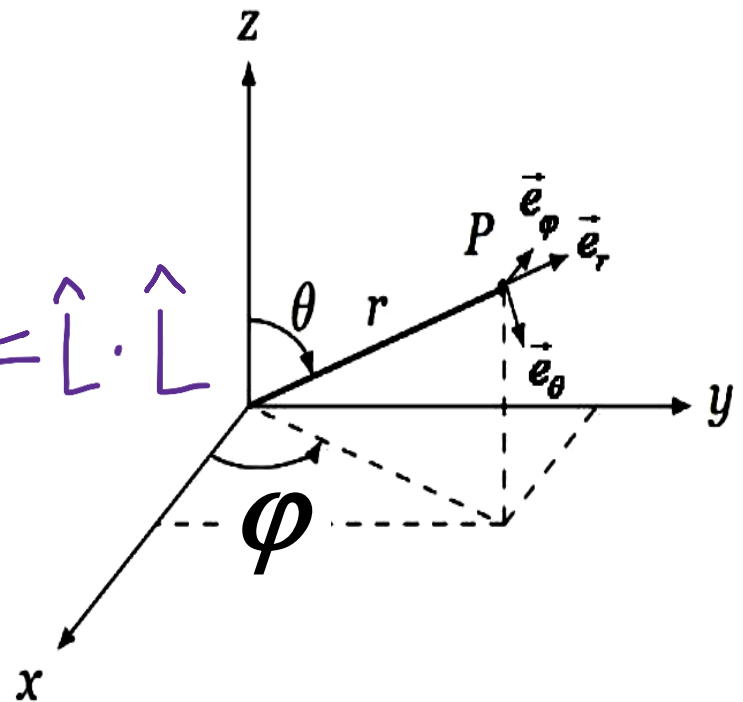
$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi},$$

$$\hat{L}_x = i\hbar(\sin \varphi \frac{\partial}{\partial \theta} + \operatorname{ctg} \theta \cos \varphi \frac{\partial}{\partial \varphi})$$

$$\hat{L}_y = -i\hbar(\cos \varphi \frac{\partial}{\partial \theta} - \operatorname{ctg} \theta \sin \varphi \frac{\partial}{\partial \varphi})$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right].$$

$$\hat{L}^2 = \hat{L} \cdot \hat{L}$$



L_z 本征函数 (重点)

★ 算符形式 $\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$,

本征方程 $\hat{L}_z \psi_m = \lambda \psi_m$

令 $\lambda = m\hbar$ (m 无量纲), 则

$$-i\hbar \frac{d}{d\varphi} \psi_m = m\hbar \psi_m(\varphi) \Rightarrow \psi_m(\varphi) = C \exp(im\varphi)$$

由单值性 $\psi_m(\varphi + 2\pi) = \psi_m(\varphi), \Rightarrow m = 0, \pm 1, \pm 2, \dots$

角动量 L_z 量子化★

归一化 $\int_0^{2\pi} |\psi_m(\varphi)|^2 d\varphi = 1,$ $\rightarrow$ 取 $C = \frac{1}{\sqrt{2\pi}}.$

$|C|^2 \cdot 2\pi = 1$

结论

本征方程 $\hat{L}_z \psi_m = \lambda \psi_m$ 导出

$$\lambda = m\hbar \quad (m = 0, \pm 1, \pm 2, \dots)$$

$$\psi_m(\phi) = \frac{1}{\sqrt{2\pi}} \exp(im\phi)$$

空间任意一个方向都可以取为z方向，所以角动量任意一个分量都是量子化的（但每个分量有不同的本征函数系）。

$\hat{L}^2$ 本征函数

- 设 $Y(\theta, \varphi)$ 是所要求的本征函数，则本征方程为

$$\hat{L}^2 Y = \lambda \hbar^2 Y \quad (\lambda \text{ 无量纲})$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right].$$

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \varphi^2} = -\lambda Y(\theta, \varphi).$$

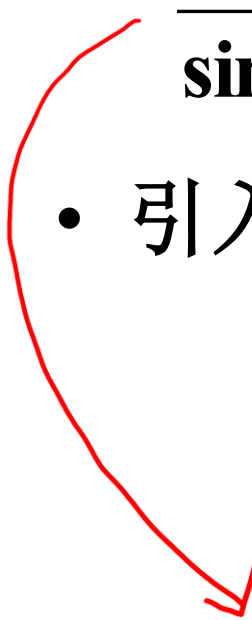
我们希望所求 $\hat{L}^2$ 的本征函数也是 $\hat{L}_z$ 的本征函数，于是设（分离变量形式的特解）

$$Y(\theta, \varphi) = P(\theta) \exp(im\varphi),$$

- 于是得方程

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dP}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} P(\theta) = -\lambda P(\theta).$$

- 引入 $w = \cos \theta, \quad w \in [-1, +1]$


$$\frac{d}{d\theta} = -\sin \theta \frac{d}{dw}$$

$$\frac{d}{dw} \left[(1-w^2) \frac{dP}{dw} \right] + \left(\lambda - \frac{m^2}{1-w^2} \right) P(w) = 0.$$

$w = \pm 1$ 是方程的奇点

下面级数求解

*方程的级数求解

$$\frac{d}{dw} \left[(1-w^2) \frac{dP}{dw} \right] + \left(\lambda - \frac{m^2}{1-w^2} \right) P(w) = 0.$$

(1) $m = 0$ (称Legendre方程)

$$\frac{d}{dw} \left[(1-w^2) \frac{dP}{dw} \right] + \lambda P = 0$$

$$(1-w^2) \frac{d^2 P}{dw^2} - 2w \frac{dP}{dw} + \lambda P = 0$$


$$P = \sum_{k=0}^{\infty} c_k w^k$$

*

$$P = \sum_{k=0}^{\infty} c_k w^k$$

$$c_{l+2} = \frac{l(l+1) - \lambda}{(l+1)(l+2)} c_l$$

$$\ln(1 - w^2) = - \sum_{n=1}^{\infty} \frac{w^{2n}}{n} \quad (\text{在 } w = 0 \text{ 附近的泰勒展开})$$

$$l \rightarrow \infty, c_{l+2} / c_l \rightarrow 1 - 2 / l,$$

当 $|w| \rightarrow 1$, P 以与 $\ln(1 - w^2)$ 相同的速度发散。

避免发散的条件是 $\lambda = l(l+1), l = 0, 1, 2, \dots$

$$P = c_0 + c_2 w^2 + c_4 w^4 + \dots \text{或 } P = c_1 w + c_3 w^3 + c_5 w^5 + \dots$$

$$P = \frac{1}{2^l l!} \frac{d^l}{dw^l} (w^2 - 1)^l \equiv P_l(w)$$

$P_l(w)$ 称为 *Legendre* 多项式

$$w = \cos \theta, \quad w \in [-1, +1]$$

(2) $m \neq 0$,

$$\frac{d}{dw} \left[(1-w^2) \frac{dP}{dw} \right] + \left(\lambda - \frac{m^2}{1-w^2} \right) P(w) = 0.$$

在奇点 $w = \pm 1$ 处一般是发散的，级数求解方程，
同样发现只有 $\lambda = l(l+1)$ 时才有物理解，为

$$P(w) = \frac{1}{2^l l!} (1-w^2)^{|m|/2} \frac{d^{l+|m|}}{dw^{l+|m|}} (w^2 - 1)^l$$

(注意 m 带绝对值)

$$l = |m|, |m| + 1, \dots \quad (|m| = 0, 1, 2, \dots, l)$$

最后的本征函数

- 球谐函数 (L^2 和 L_z 共同本征函数)

$$Y_{lm}(\theta, \varphi) = N_{lm} P_l^m(\cos \theta) \exp(im\varphi),$$

归一化系数

$$\int Y_{lm}^*(\theta, \varphi) \cdot Y_{lm}(\theta, \varphi) \cdot d\Omega = 1. \quad (d\Omega = \sin \theta \cdot d\theta d\varphi) \Rightarrow$$

$$N_{lm} = (-1)^m \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}.$$

(教材 3.2.21 中的 m 少了绝对值符号) 更正

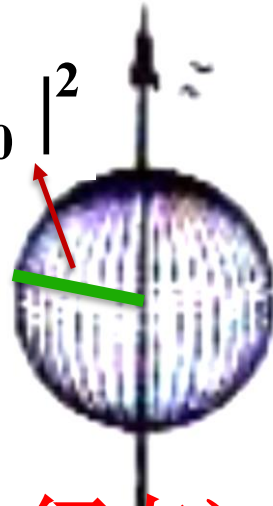
球谐函数 (L^2 和 L_z 共同本征态)

举例:

1. $l=0$:

$$Y_{00} = \sqrt{\frac{1}{4\pi}},$$

$$|Y_{00}|^2$$



(无方向性, 也是角动量其它分量的本征态)

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$$

$$\hat{L}_z Y_{00} = 0$$

$$\hat{L}_x = i\hbar (\sin \varphi \frac{\partial}{\partial \theta} + \cot \theta \cos \varphi \frac{\partial}{\partial \varphi})$$

$$\hat{L}_x Y_{00} = 0$$

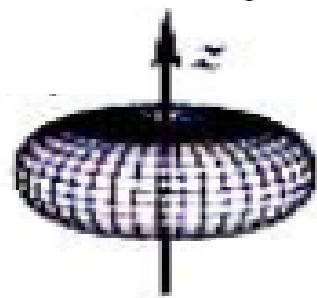
$$\hat{L}_y = -i\hbar (\cos \varphi \frac{\partial}{\partial \theta} - \cot \theta \sin \varphi \frac{\partial}{\partial \varphi})$$

$$\hat{L}_y Y_{00} = 0$$

$$2. \quad l=1 \quad Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta, \quad |Y_{10}|^2$$



$$Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta \exp(\pm i\varphi).$$



(L^2 和 L_z 共同本征态)

注意：以上是 L_z 本征态，不是 L_x 和 L_y 本征态，但 L_z 本征函数可以构成完全函数系（即可以线性组合生成包括 L_x 和 L_y 本征态在内的所有角度空间的波函数 $\psi(\theta, \phi)$ ）。

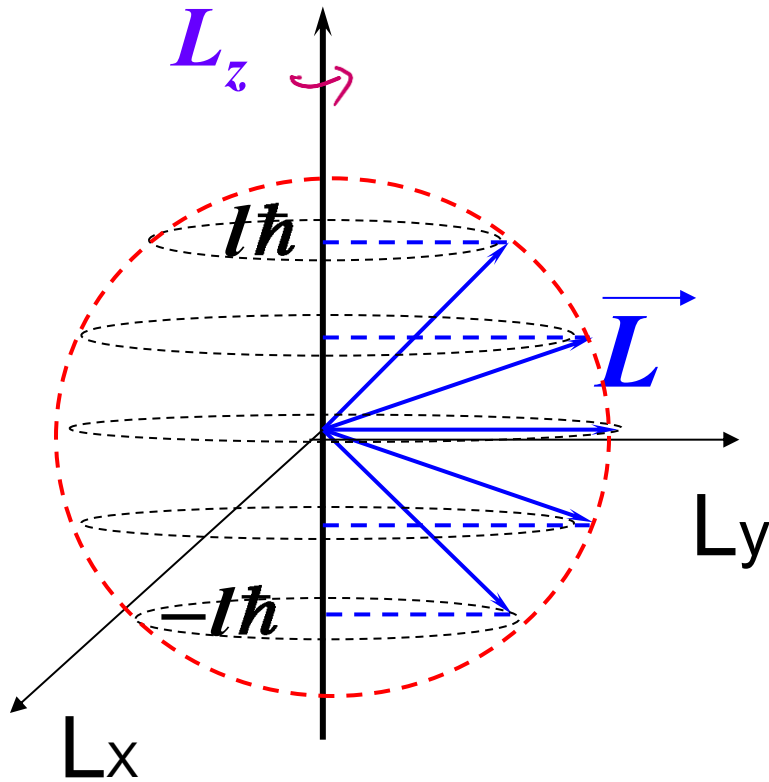
球谐函数的基本性质

$$(1) \quad \left\{ \begin{array}{l} \hat{L}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm}, \quad l = 0, 1, 2, \dots, \\ \hat{L}_z Y_{lm} = m\hbar Y_{lm}, \quad m = l, l-1, \dots, -l. \end{array} \right. \quad (\text{重点})$$

$$(2) \quad \int Y_{l'm'}^*(\theta, \varphi) \cdot Y_{lm}(\theta, \varphi) \cdot d\Omega = \delta_{l'l} \delta_{m'm}. \quad (2l+1 \text{ 个值})$$

$(d\Omega = \sin \theta \cdot d\theta d\varphi)$ Y_{lm} 的正交归一性。

$$(3) \quad Y_{lm}^*(\theta, \varphi) = (-1)^m Y_{l, -m}(\theta, \varphi).$$



注意 L_z 的上限是 $l\hbar$
 而不是 $\sqrt{l(l+1)} \hbar^2$
 (不是经典图像)

$Y_{lm}(\theta, \phi)$ 态下, L^2 和 L_z^2 都有确定的值,

$$L_x^2 + L_y^2 = L^2 - L_z^2 = l(l+1) \hbar^2 - m^2 \hbar^2$$

(除了 $Y_{00}(\theta, \phi)$ 态外, 总有 $L_x^2 + L_y^2 > 0$)

需要记住的结果：

$$l = \underline{0}, 1, 2, \dots; \quad \rightarrow L^2 = l(l+1)\hbar^2$$

$$m = l, l-1, \dots, -l. \quad \rightarrow L_z = m\hbar$$

↓

$$\hat{L}^2 \text{ 的简并度: } 2l+1$$

（一个本征值对应k个线性独立的本征函数（k=2,3,4.....）的现象叫做简并，k称为简并度）

原子物理的术语

- 态的记号

$$l = 0, 1, 2, 3, \dots$$

S, P, D, F态

作业

一. 教材作业 p.92, #3.5 (刚性转子), 截图如下:

3.5 一刚性转子转动惯量为 I , 它的能量的经典表示式是 $H = \frac{L^2}{2I}$, L 为角动量,

求与此对应的量子体系在下列情况下的定态能量及波函数:

(1) 转子绕一固定轴转动:

(2) 转子绕一固定点转动:

二. 补充题3.2: 直接证明 $Y_{00}, Y_{10}, Y_{11}, Y_{1,-1}$ 是正交归一的 (利用以下结果)

$$Y_{00} = \sqrt{\frac{1}{4\pi}}, \quad Y_{10} = \sqrt{\frac{3}{4\pi}} \cos \theta,$$

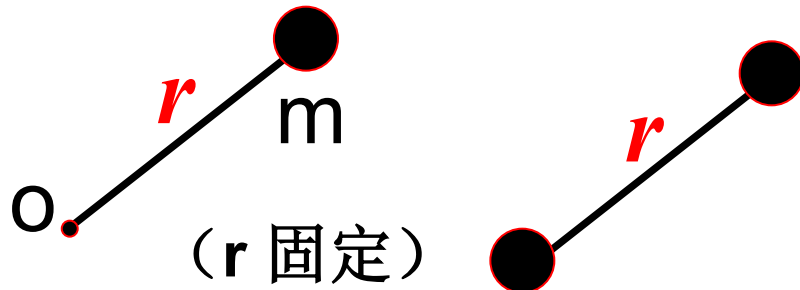
$$Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta \exp(\pm i\varphi).$$

$$\begin{aligned} &\rightarrow \int Y_{00}^* Y_{10} d\Omega \\ &= 0 \\ &\int Y_{00}^* Y_{00} d\Omega = 1 \\ &\dots \end{aligned}$$

(提醒: 作业补充题是要求完成的)

关于刚性转子（作业3.5中用到）

$$H = \frac{L^2}{2I}$$



可以推广到双
原子分子的转
动, m 用约化质
量 μ 代替。

【简单解释：考虑质点 m 被轻杆连接，绕 O 点转动，则

经典力学中 $L = p r$ ，于是 $H = \frac{p^2}{2m} = \frac{(L / r)^2}{2m} = \frac{L^2}{2I} (I = m r^2)$ 】

过渡到量子力学， L^2 本征值是量子化的，于是 $H (= \frac{L^2}{2I} = \frac{L^2}{2\mu r^2})$

本征值也是量子化的。（注意：对一般转子， L^2 本征值仍然是 $l(l+1)\hbar^2$ ，但对限制在 xy 平面中运动的转子，角动量只有一个 L_z 分量，且 L_z 本征值是 $m_l \hbar$ ($m_l = 0, \pm 1, \pm 2, \dots$) 。

§ 3.3 中心力场中的运动，氢原子

1. 模型的建立

2. 求解氢原子（内部）相对运动

3. 波函数

1. 模型的建立

1.1 经典模型:

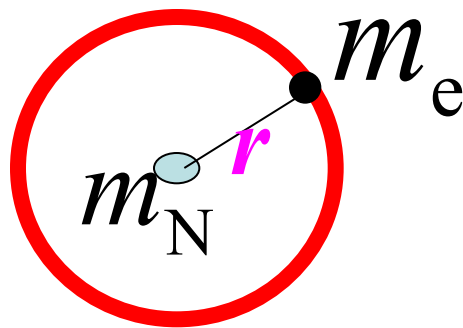
$$H = \frac{1}{2} m_N \dot{\vec{r}}_N^2 + \frac{1}{2} m_e \dot{\vec{r}}_e^2 + U(|\vec{r}_e - \vec{r}_N|) \quad (1)$$

$$\vec{r} = \vec{r}_e - \vec{r}_N \text{ (相对坐标)}$$

引入质心坐标 $\vec{R}$ 和相对坐标 $\vec{r}$

$$\begin{cases} \vec{R} = \frac{m_N \vec{r}_N + m_e \vec{r}_e}{m_N + m_e} \\ \vec{r} = \vec{r}_e - \vec{r}_N \end{cases} \Rightarrow \begin{cases} \vec{r}_N = \vec{R} - \frac{m_e}{m_e + m_N} \vec{r} \\ \vec{r}_e = \vec{R} + \frac{m_N}{m_e + m_N} \vec{r} \end{cases}$$

代入 (1)
 $\dot{\vec{R}} \cdot \dot{\vec{r}}$ 项抵消
得下页结果



$$\mathbf{H} = \frac{1}{2} \mathbf{M} \dot{\mathbf{R}}^2 + \frac{1}{2} \mu \dot{\mathbf{r}}^2 + \mathbf{U}(\mathbf{r})$$

(质心运动) (相对运动)

定义 $\vec{p}_R = \mathbf{M} \dot{\mathbf{R}}$; $\vec{p} = \mu \dot{\mathbf{r}}$

$$\mathbf{H} = \frac{\vec{p}_R^2}{2\mathbf{M}} + \frac{\vec{p}^2}{2\mu} + U(r)$$

(质心运动) (相对运动)

$$M = m_N + m_e$$

$$\mu = \frac{m_N m_e}{M} \text{ (约化质量)}$$

$\vec{p} = \mu \dot{\vec{r}}$ 的物理含义

$$\left\{ \begin{array}{l} \vec{r}_N = \vec{R} - \frac{m_e}{m_e + m_N} \vec{r} \Rightarrow \text{质心系中原子核坐标} \\ \vec{r}_N' = \vec{r}_N - \vec{R} = -\frac{m_e}{m_e + m_N} \vec{r}, \\ \vec{r}_e = \vec{R} + \frac{m_N}{m_e + m_N} \vec{r} \Rightarrow \text{质心系中电子坐标} \\ \vec{r}_e' = \vec{r}_e - \vec{R} = \frac{m_N}{m_e + m_N} \vec{r} \end{array} \right.$$

$$\text{质心系中电子动量 } \vec{p}_e' = m_e \dot{\vec{r}}_e' = \frac{m_e m_N}{m_e + m_N} \dot{\vec{r}} = \mu \dot{\vec{r}} = \vec{p}$$

$$\text{质心系原子核动量 } \vec{p}_N' = m_N \dot{\vec{r}}_N' = -\frac{m_e m_N}{m_e + m_N} \dot{\vec{r}} = -\mu \dot{\vec{r}} = -\vec{p}$$

$$\frac{p^2}{2\mu} = \left(\frac{1}{m_e} + \frac{1}{m_N} \right) \frac{p^2}{2} = \frac{p^2}{2m_e} + \frac{p^2}{2m_N}$$

(质心系中电子和原子核的总动能)

类似地， $\vec{L} = \vec{r} \times \vec{p}$ 的物理含义是质心参考系中电子和原子核绕质心的总角动量
(证明留作思考题)

附：思考题解答：

前面已经证明了 $\vec{p}$ 代表质心系中电子动量或原子核动量的相反数，即 $\vec{p} = \vec{p}_e' = -\vec{p}_N'$ ，

$$\text{所以 } \vec{L} = \vec{r} \times \vec{p} = (\vec{r}_e - \vec{r}_N) \times \vec{p} = \underline{(\vec{r}_e' - \vec{r}_N')} \times \vec{p}$$

$$= \vec{r}_e' \times \vec{p}_e' - \vec{r}_N' \times \vec{p}_e'$$

$$= \vec{r}_e' \times \vec{p}_e' - \vec{r}_N' \times (-\vec{p}_N')$$

$$= \underline{\vec{r}_e' \times \vec{p}_e'} + \underline{\vec{r}_N' \times \vec{p}_N'}$$

$$= \underline{\vec{L}_e'} + \underline{\vec{L}_N'}$$

(即质心系中电子和原子核总角动量)

1.2 量子模型:

$$\text{量子化: } \vec{p}_R = -i\hbar\nabla_R$$

$$\text{量子化: } \vec{p} = -i\hbar\nabla$$

$$H = H = \frac{p_R^2}{2M} + \frac{p^2}{2\mu} + U(r)$$

(质心运动) (相对运动)

$$= -\frac{\hbar^2}{2M}\nabla_R^2 - \frac{\hbar^2}{2\mu}\nabla^2 + U(r)$$

$$(\nabla_R^2 = \frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2}; \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2})$$

$$i\hbar \frac{\partial}{\partial t} \Psi(\vec{r}, \vec{R}, t) = H\Psi$$

求解薛定谔方程:

设分离变量特解: $\Psi(\vec{r}, \vec{R}, t) = \tilde{\Psi}(\vec{r}, \vec{R}) e^{-\frac{i}{\hbar} E_{\text{总}} t}$

定态薛定谔方程 $H \tilde{\Psi}(\vec{r}, \vec{R}) = E_{\text{总}} \tilde{\Psi}(\vec{r}, \vec{R})$

$$\left[-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 - \frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \tilde{\Psi}(\vec{r}, \vec{R}) = E_{\text{总}} \tilde{\Psi}(\vec{r}, \vec{R}) \quad (2)$$

继续分离变量: 设 $\tilde{\Psi}(\vec{r}, \vec{R}) = \psi(\vec{r}) \phi(\vec{R})$, 代入(2)得

$$\left[-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 - \frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi(\vec{r}) \phi(\vec{R}) = E_{\text{总}} \psi(\vec{r}) \phi(\vec{R})$$

两边同除以 $\psi(\vec{r}) \phi(\vec{R})$,

$$\frac{-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 \phi(\vec{R})}{\phi(\vec{R})} + \frac{-\frac{\hbar^2}{2\mu} \nabla^2 \psi(\vec{r})}{\psi(\vec{r})} + U(r) = E_{\text{总}}$$

↪ E

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi(\vec{r}) = E \psi(\vec{r}) \text{ (相对运动)}$$

(这里只关心相对运动)
(原子内部运动)

$$-\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 \phi(\vec{R}) = (E_{\text{总}} - E) \phi(\vec{R}) \text{ (质心运动)}$$

自由

2. 求解氢原子（内部）相对运动

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi = E \psi(\vec{r}),$$

动能



$$-\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{2\mu r^2} \hat{L}^2$$

转动动能

解释如下

附：角动量平方和拉普拉斯算符的关系

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right].$$
$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}$$

思考题：动能是量子化的吗？

$$\rightarrow -\frac{\hbar^2}{2\mu} \nabla^2 = -\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{2\mu r^2} \hat{L}^2$$

答：否

求解相对运动方程

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) \right] \psi = E \psi(\vec{r}),$$

动能



$$\left[-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + U(r) + \frac{1}{2\mu r^2} \hat{L}^2(\theta, \phi) \right] \psi = E \psi(\vec{r}).$$

径向动能

转动动能

移项, 两边同乘 r^2

$$\left[-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + U(r) - E \right] \psi r^2 = -\frac{1}{2\mu} \hat{L}^2(\theta, \phi) \psi$$

径向

角向

$$\left[-\frac{\hbar^2}{2\mu r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + U(r) - E \right] \psi r^2 = -\frac{1}{2\mu} \hat{L}^2(\theta, \phi) \psi$$

两边同除以 $R(r)Y(\theta, \phi)$

得

$$\left[\frac{-\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right)}{R(r)} + U(r) - E \right] r^2 = -\frac{\frac{1}{2\mu} \hat{L}^2 Y(\theta, \phi)}{Y(\theta, \phi)}$$

↓ 径向

角向

两边必须等于同一个常数，设为 $-\frac{\lambda \hbar^2}{2\mu}$ ，
得到下面两个方程：

$$\left\{ \begin{aligned} \frac{\hat{L}^2 Y(\theta, \phi)}{Y(\theta, \phi)} &= \lambda \hbar^2 \left[\text{前面级数求解得 } \lambda = l(l+1)\hbar^2 \right] \quad (1) \\ -\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + U(r) - E &= -\frac{\lambda \hbar^2}{2\mu r^2} \quad (2) \end{aligned} \right.$$

$$-\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[U(r) + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] R = ER(r).$$

(径向方程)

“离心势能” (转动动能)

库仑势 $U(r) = -k_1 \frac{Ze^2}{r}$

(氢原子 $Z=1$)

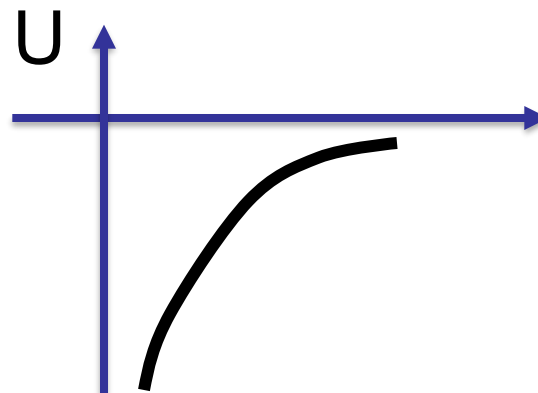
$$k_1 = \begin{cases} (4\pi\epsilon_0)^{-1}, & (\text{SI}) \\ 1, & (\text{CGS}) \end{cases}$$

再引入变换 $R(r) = \frac{u(r)}{r}$ ($r \rightarrow 0$ 时 $u \rightarrow 0$)

得
$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[-k_1 \frac{Ze^2}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] u = Eu(r).$$

以下只分析束缚态,

$$E < U(\infty) = 0$$



$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[-k_1 \frac{Ze^2}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} \right] u = \underline{Eu(r)}.$$

• 定义 $\rho = \alpha r$ $\left(\alpha = \frac{\sqrt{8\mu|E|}}{\hbar} \right),$

无量纲化

$$\beta = \frac{2\mu k_1 Ze^2}{\alpha \hbar^2} = \frac{k_1 Ze^2}{\hbar} \sqrt{\frac{\mu}{2|E|}}$$

$$\frac{d^2 u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2} \right] u = 0. \quad (1)$$

(此处 $E < 0$, 否则没有束缚态解)

两种极限情况

$\rho \rightarrow \infty$, 保留 (1) 的[...]中第1项

$$\frac{d^2 u}{d\rho^2} - \frac{1}{4} u \approx 0, \quad u(\rho) \rightarrow \exp\left(-\frac{1}{2}\rho\right);$$

$\rho \rightarrow 0$ (保留 (1) 的[...]中第3项)

$$\frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u \approx 0, \quad u(\rho) \rightarrow \rho^{l+1}.$$

一般解

设 $u(\rho) = \overset{\rho \rightarrow 0}{\downarrow} \rho^{l+1} \exp\left(\overset{\rho \rightarrow \infty}{\downarrow} -\frac{1}{2}\rho\right) \underline{v(\rho)},$

代入方程

$$\frac{d^2 u}{d\rho^2} + \left[-\frac{1}{4} + \frac{\beta}{\rho} - \frac{l(l+1)}{\rho^2} \right] u = 0.$$

得

$$\frac{d^2 v}{d\rho^2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \frac{dv}{d\rho} + \frac{\beta - (l+1)}{\rho} v = 0$$

求解 $\frac{d^2 v}{d\rho^2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \frac{dv}{d\rho} + \frac{\beta - (l+1)}{\rho} v = 0$

设 $v = \sum_{k=0}^{\infty} c_k \rho^k$, 代入以上方程, 得

$$\sum_{k=2}^{\infty} c_k k(k-1) \rho^{k-2} + \left[\frac{2(l+1)}{\rho} - 1 \right] \sum_{k=1}^{\infty} c_k k \rho^{k-1} + \frac{\beta - (l+1)}{\rho} \sum_{k=0}^{\infty} c_k \rho^k = 0,$$

比较 ρ 的同类项系数，得

$$c_{k+1} = \frac{k - [\beta - (l + 1)]}{(k + 1)k + 2(k + 1)(l + 1)} c_k$$

截断条件：必须存在某个 n_r ，取 $k=n_r$ 时 $c_{n_r+1} = 0$
عزوة

(之后的系数当然也就为0) ，

这要求： $n_r - [\beta - (l + 1)] = 0$

造成能量量子化！

$$\beta = n_r + l + 1 \equiv n, \quad (n_r = 0, 1, 2, \dots)$$

$$\beta = \frac{k_1 Z e^2}{\hbar} \sqrt{\frac{\mu}{2|E|}} = n \quad (\text{氢原子 } Z=1)$$

$$= n_r + l + 1 (n_r = 0, 1, 2, \dots)$$

$$l = n - n_r - 1 \Rightarrow l = 0, 1, \dots, n - 1$$

得到能级 $E_n = -\frac{\mu k_1^2 Z^2 e^4}{2\hbar^2} \frac{1}{n^2} (n = 1, 2, 3, \dots)$

